

2- The mathematical model relating the changes of an internal combustion engine speed Δw with the position of the fuel rack Δz is given by:

$$2 \frac{dw}{dt} + 0.2 \Delta w = \Delta z(t)$$

Find:

- 1- The unit impulse response of the engine
- 2- The unit step response of the engine
- 3- The Engine response to the signal shown in fig 2

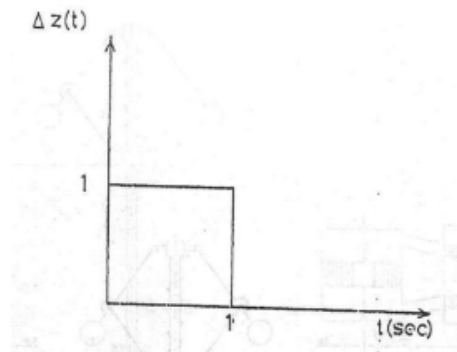


Fig 2

$$\frac{w(s)}{z(s)} = \frac{1}{2s + 0.2} = \frac{5}{10s + 1}$$

$$1) \quad \frac{5}{10} e^{-t/10}$$

$$2) \quad 5(1 - e^{-t/10})$$

$$3) \quad 5(1 - e^{-t/10}) - 5(1 - e^{-(t-1)/10}) u(t-1)$$

if the mathematical model is given by:

$$2 \frac{dw}{dt} + 0.2 \Delta w = \Delta z(t-1)$$

Find

4- The unit impulse response of the engine

5- The unit step response of the engine

$$\frac{w(s)}{z(s)} = \frac{5}{10s+1} e^{-1}s$$

$$4) \quad 5/10 e^{-1/10(t-1)} u(t-1)$$

$$5) \quad 5(1 - e^{-1/10(t-1)}) u(t-1)$$

- 3- Drive the mathematical models relating the head (h) in the accumulator to the input (Q) shown in fig (3). Also, the relation between (Q_1) and (Q_2) in the transmission system shown in Fig. (4).

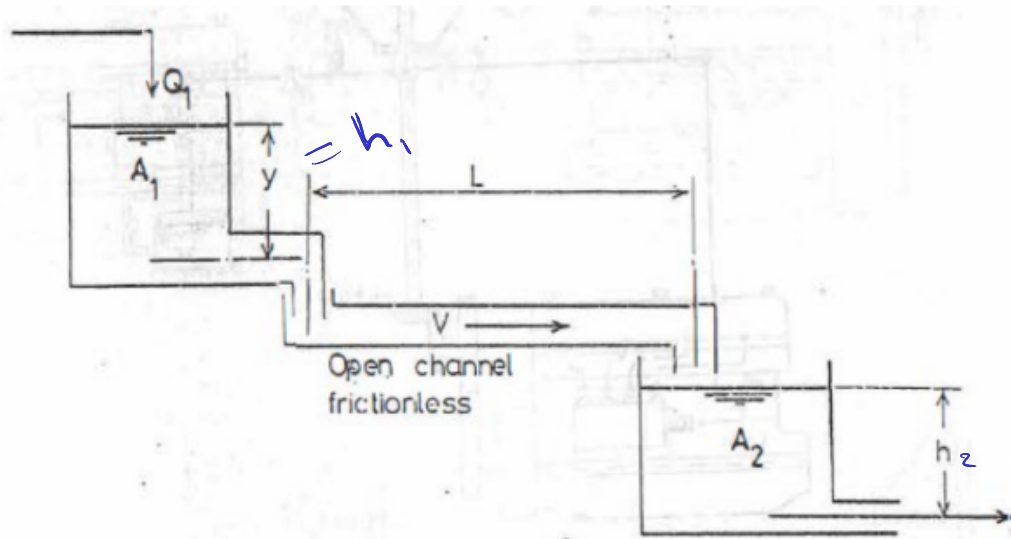


Fig. 3

$$\frac{H_2(s)}{Q_1(s)} = \frac{H_1(s)}{Q_1(s)} \cdot \frac{H_2(s)}{H_1(s)}$$

$$A_1 s H_1(s) = Q_1(s) - C_1 H_1(s)$$

$$\frac{H_1(s)}{Q_1(s)} = \frac{1}{A_1 s + C_1}$$

$$A_2 s H_2(s) = C_1 H_1(s) - C_2 H_2(s)$$

$$\frac{H_2(s)}{H_1(s)} = \frac{C_1}{A_2 s + C_2}$$

$$\frac{H_2(s)}{Q_1(s)} = \frac{C_1}{(A_1 s + C_1)(A_2 s + C_2)}$$

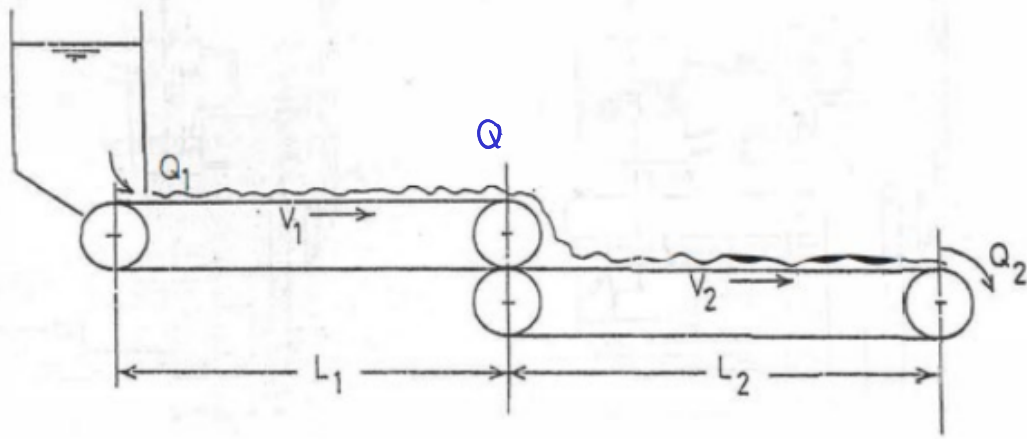


Fig. 4

$$Q(t - \tau_1) = Q(t) \quad \tau_1 = \frac{L_1}{V_1}$$

$$Q(t - \tau_2) = Q_2(t) \quad \tau_2 = \frac{L_2}{V_2}$$

4- Fig (5) shows a hydraulic tank with its input and output valves.

Find:

- The mathematical model relating the discharge q_i to the tank and the positions of the input and output valves Δu , $\Delta \lambda$.
- The mathematical model relating the head y and the signals Δu , $\Delta \lambda$.
- The impulse and unit-step responses of the tank in cases a and b.

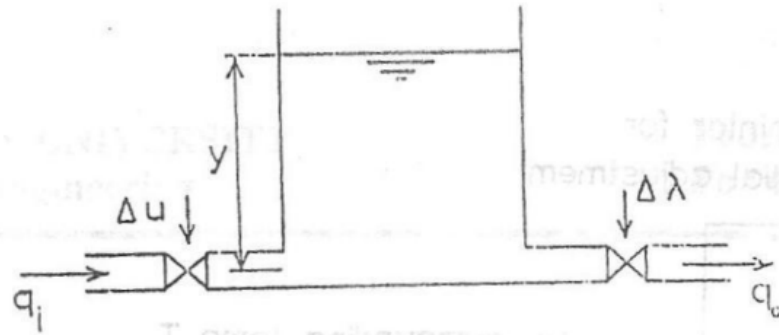


Fig. 5

$$AsY(s) = c_1 U(s) - c_2 Y(s) - c_3 \lambda(s) + c_4 Y(s)$$

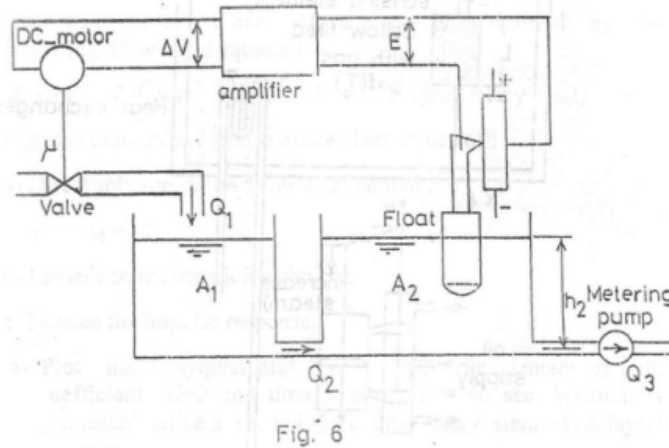
$$Y(s) = \frac{c_1}{As + \underbrace{c_2 + c_4}_{c_5}} U(s) - \frac{c_3}{As + \underbrace{c_2 + c_4}_{c_5}} \lambda(s)$$

$$y(t) = \frac{c_1}{c_5} \left(1 - e^{-\frac{c_5}{A}t} \right) - \frac{c_3}{c_5} \left(1 - e^{-\frac{c_5}{A}t} \right)$$

5- Fig (6) shows a plant of two hydraulic tanks with a metering pump and the level (h_2) is controlled via the system shown in the Figure.
Find:

- The plant transfer function relating the head (h_2) with the variation of (Q_1) if (Q_3) is constant.
- Draw the block diagram of the system showing all signals in the Figure and also the transfer functions of the different components.
- Find the unit- step response of the plant if it has a transfer function in the form

$$W(s) = \frac{10 e^{-5s}}{s(50s + 1)}$$



step:
$$\frac{10}{s^2(50s+1)} = \frac{As+B}{s^2} + \frac{C}{50s+1}$$

$$(As+B)(50s+1) + Cs^2 = 10$$

$$50As^2 + As + 50Bs + B + Cs^2 = 10$$

$$s^0: B = 10$$

$$s^1: As + 50B = 0 \rightarrow A = -500$$

$$s^2: 50A + C = 0 \rightarrow C = 25000$$

$$\begin{aligned} & -500 + 10t + 500 e^{-1/50t} \\ & \left[-500 + 10(t-5) + 500 e^{-1/50(t-5)} \right] u(t-5) \end{aligned}$$

6- Referring to the system shown in Figure 7, determine the values of K and k such that the system has a damping ratio ζ of 0.7 and an undamped natural frequency ω_n of 4 rad/sec.

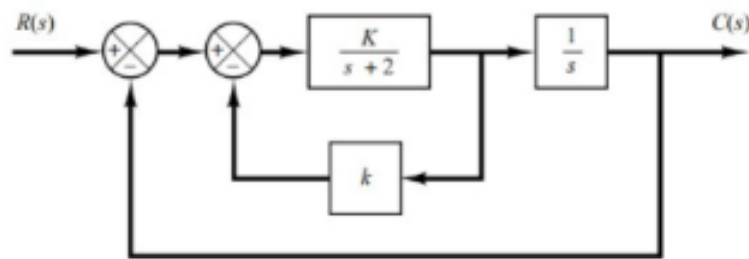


Fig 7.

$$\frac{1}{s} \cdot \frac{\frac{K}{s+2}}{1 + \frac{Kk}{s+2}} = \frac{K}{s(s+2 + Kk)}$$

$$\frac{K}{s^2 + (2 + Kk)s + K}$$

$$2 + Kk = 2\zeta\omega_n$$

$$K = \omega_n^2$$

$$K = 16$$

$$\frac{2(0.7)(4) - 2}{16} = 12$$